

ROSTOV-ON-DON, Russian Federation

WMO: 347300

Lat: 47.258N

Lon: 39.818E

Elev: 79

StdP: 100.38

Time Zone: 3.00 (E03)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-14.9	-21.1	0.6	-16.8	-18.2	0.8	-14.3	14.2	-8.4	13.1	-6.7	5.0	80	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.1	21.1	33.1	20.8	31.2	20.1	23.3	31.1	22.3	29.7	21.3	28.4	5.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	15.8	26.9	19.9	14.8	25.7	19.0	13.9	25.1	69.6	31.3	65.8	29.7	62.6	28.3	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.3	10.7	9.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-19.8	37.3	3.9	1.9	-22.6	38.7	-24.9	39.8	-27.1	40.9	-29.9	42.3		
			-20.1	24.7	3.9	1.4	-22.9	25.7	-25.2	26.5	-27.3	27.3	-30.1	28.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-3.1	-2.0	3.5	10.9	17.2	21.6	24.3	24.0	17.9	10.2	3.8	-1.0
	DBStd	10.85	6.08	5.87	4.40	3.88	3.98	3.49	3.16	3.59	4.02	4.72	4.91	5.46
	HDD10.0	1575	406	336	206	35	2	0	0	0	2	56	192	341
	HDD18.3	3347	664	569	461	224	70	9	1	2	56	253	437	599
	CDD10.0	1822	0	0	3	62	225	348	444	433	237	63	6	0
	CDD18.3	552	0	0	0	1	35	108	187	177	41	3	0	0
	CDH23.3	5429	0	0	0	11	312	980	1921	1868	316	21	0	0
	CDH26.7	2228	0	0	0	2	76	356	846	859	86	4	0	0
Wind	WSAvg	4.3	4.8	5.0	5.1	4.7	3.8	3.6	3.6	3.7	4.1	4.1	4.4	4.6
Precipitation	PrecAvg	593	54	41	38	47	60	60	55	42	41	33	51	72
	PrecMax	801	129	92	92	128	150	147	166	116	171	107	129	151
	PrecMin	426	6	2	1	5	5	9	2	4	0	2	6	4
	PrecStd	93	32	22	20	26	34	32	39	30	34	22	34	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	12.2	17.9	24.9	31.6	34.2	37.2	38.1	32.0	26.2	17.2	11.1
		MCWB	7.1	8.4	11.0	15.0	21.4	20.8	22.4	21.1	20.6	17.8	12.7	8.7
	2%	DB	6.9	8.8	14.2	21.9	28.8	32.2	35.2	35.2	29.0	22.0	14.2	9.0
		MCWB	5.7	6.5	8.9	13.6	18.6	20.8	21.3	20.9	18.5	14.9	11.3	7.8
	5%	DB	5.0	6.6	12.0	19.8	26.8	30.6	33.1	33.2	27.0	19.2	12.0	7.1
		MCWB	4.2	5.1	7.6	12.6	17.7	20.0	21.2	20.2	17.8	14.2	9.9	6.1
	10%	DB	3.8	4.8	9.9	17.9	24.9	28.8	31.1	31.2	24.9	17.0	10.1	5.8
		MCWB	3.1	3.5	6.7	11.7	17.1	19.4	20.2	19.7	17.0	13.3	8.7	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	9.4	11.3	15.7	22.3	23.3	25.1	25.0	21.8	18.4	13.3	9.5
		MCDB	8.6	11.8	16.3	22.2	29.9	30.8	33.4	32.7	28.1	23.9	15.7	10.2
	2%	WB	5.6	6.7	9.8	14.5	19.9	22.1	23.4	23.1	20.0	16.3	11.6	7.7
		MCDB	6.4	8.0	13.5	20.2	26.3	28.9	31.3	31.0	26.1	20.4	14.1	8.8
	5%	WB	4.1	5.0	8.2	13.3	18.7	21.2	22.5	21.9	18.9	15.0	10.1	6.1
		MCDB	4.7	6.1	11.0	18.3	24.7	27.9	30.1	29.2	24.6	18.1	11.4	6.9
	10%	WB	2.8	3.3	6.8	12.3	17.5	20.4	21.4	21.0	17.8	13.6	8.8	4.8
		MCDB	3.4	4.2	9.2	16.8	23.1	26.7	28.5	28.3	23.2	16.5	10.0	5.4
Mean Daily Temperature Range	5% DB	MDBR	5.0	6.2	7.5	9.7	10.4	10.4	10.9	11.6	10.6	8.4	6.0	5.1
		MCDBR	5.5	8.1	11.4	12.7	12.9	12.6	13.1	13.7	13.1	11.7	8.4	6.5
		MCWBR	4.9	6.2	7.1	6.7	6.3	5.4	5.4	5.7	6.2	7.0	6.1	5.6
	5% WB	MCDBR	5.2	7.3	10.2	11.0	11.7	11.0	11.6	12.0	11.8	10.7	7.8	6.1
		MCWBR	4.8	6.0	6.8	6.2	6.3	5.7	5.9	6.2	6.3	6.8	6.1	5.5
Clear-Sky Solar Irradiance	taub	0.342	0.375	0.412	0.457	0.419	0.424	0.448	0.447	0.416	0.386	0.355	0.346	
	taud	2.221	2.125	2.127	2.057	2.255	2.269	2.212	2.208	2.262	2.305	2.312	2.236	
	Ebn at Noon	709	760	794	795	847	843	815	798	786	749	694	652	
	Edh at Noon	83	113	131	155	132	131	137	132	114	94	76	74	
All-Sky Solar Radiation	RadAvg	0.98	1.87	2.94	4.60	6.01	6.46	6.34	5.74	4.14	2.41	1.19	0.75	
	RadStd	0.21	0.31	0.32	0.46	0.58	0.49	0.34	0.38	0.35	0.33	0.22	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+71
(47)	Regional (9 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-150	+131	+93

Nomenclature: See separate page